

PROJECT REPORT

TITLE: Azure Firewall Centralised Egress Filtering

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Department: Computer Science and Engineering (CSE)

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Abstract

Azure Firewall Centralised Egress Filtering is a cloud network security solution that routes all outbound internet traffic from multiple Azure Virtual Networks (VNETs) through a single Azure Firewall. This centralized architecture provides a unified point for monitoring, filtering, and controlling internet-bound traffic. The firewall uses Azure Firewall Policy and FQDN (Fully Qualified Domain Name) rules to allow access only to approved websites and services. Unauthorized or malicious destinations are blocked, improving the overall security of the cloud environment. The solution enhances visibility into outbound traffic and enables centralized logging and monitoring. It also simplifies security policy management across multiple virtual networks. User-Defined Routes (UDRs) are used to redirect traffic through the firewall before it reaches the internet. This architecture helps organizations reduce the attack surface and enforce compliance with security standards. The project demonstrates the design, configuration, and testing of centralized egress filtering in Microsoft Azure. It also discusses implementation challenges such as route misconfiguration and firewall costs. Overall, the solution provides a secure, scalable, and efficient approach to managing outbound network traffic in Azure.

Objectives

The main objectives of the project are:

- Route all outbound traffic through one Azure Firewall.
- Enforce an FQDN allowlist for outbound internet access.
- Block unauthorized internet destinations.
- Centralize logging and monitoring.
- Improve network security through controlled egress traffic.

Problem Statement

In many cloud environments, virtual machines and applications access the internet directly, making it difficult to monitor outbound traffic and enforce security policies. A centralized firewall architecture provides a single control point for outbound traffic, enabling organizations to restrict internet access to approved domains and reduce the attack surface.

Use Cases

1. Centralized Outbound Traffic Control

All virtual machines in multiple Azure virtual networks send internet-bound traffic through one Azure Firewall.

2. FQDN Allowlist Enforcement

Only approved domains such as:

- windowsupdate.microsoft.com
- github.com
- azure.microsoft.com

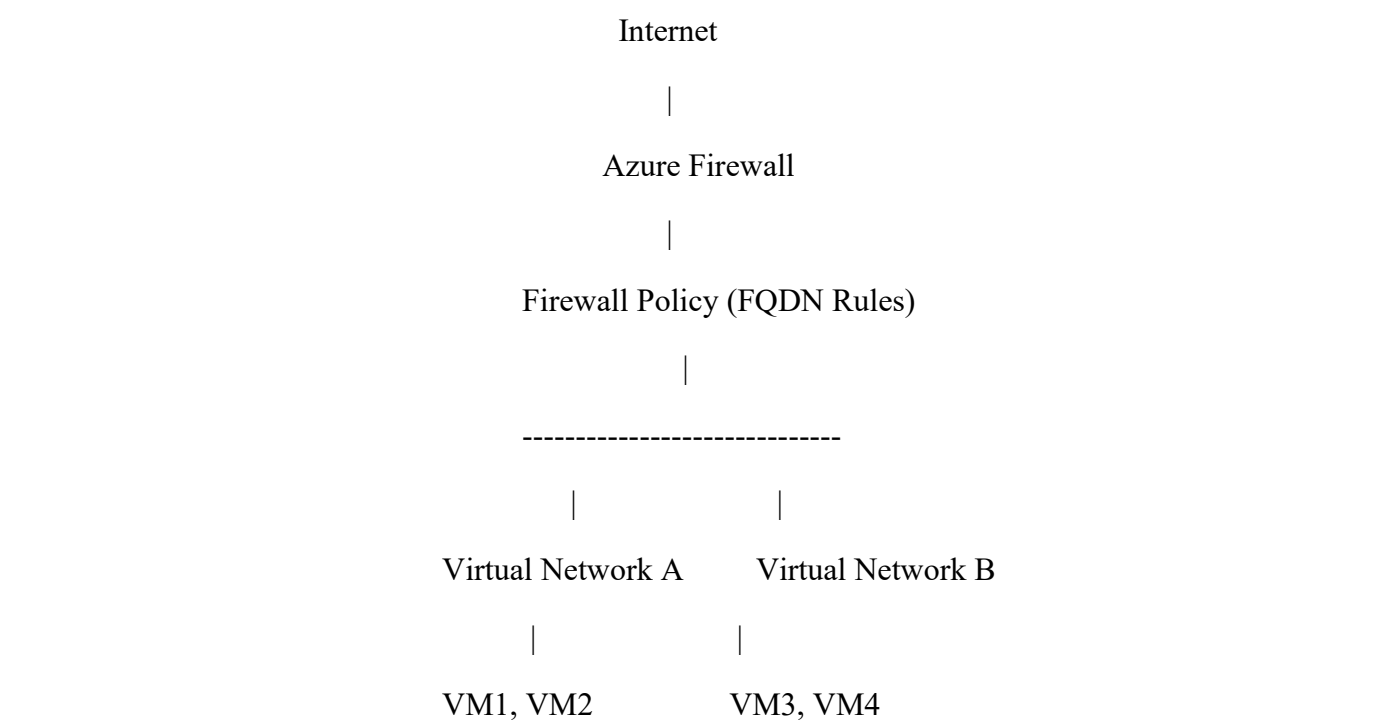
are allowed. All other domains are blocked.

3. Security Monitoring

Firewall logs provide visibility into:

- Allowed connections
- Blocked connections
- Source and destination information
- Application access attempts

System Architecture



All outbound traffic from both VNets is routed through Azure Firewall.

Azure Services Used

Service	Purpose
Azure Virtual Network	Network isolation
Azure Firewall	Centralized egress filtering
Azure Firewall Policy	Rule management
Route Tables (UDR)	Traffic redirection
Azure Monitor	Logging and monitoring
Log Analytics Workspace	Firewall log storage

Implementation Steps

Step 1: Create Virtual Networks

- VNet-A
- VNet-B

Deploy virtual machines in both networks.

Step 2: Deploy Azure Firewall

- Deploy Azure Firewall in a dedicated subnet named:
- AzureFirewallSubnet
- Assign a public IP address.

Step 3: Create Firewall Policy

Create a Firewall Policy and configure:

Application Rules

Allow:

- github.com
- azure.microsoft.com
- windowsupdate.microsoft.com

Deny all other FQDNs.

Step 4: Configure User-Defined Routes

Create a route table.

Add route:

- Address Prefix: 0.0.0.0/0
- Next Hop Type: Virtual Appliance
- Next Hop IP: Azure Firewall private IP

Associate the route table with application subnets.

Step 5: Test Connectivity

- Allowed domains are reachable.
- Blocked domains are denied.

Firewall Rule Configuration

Application Rule Collection

Rule	FQDN	Action
AllowAzure	azure.microsoft.com	Allow
AllowGitHub	github.com	Allow
AllowWindowsUpdate	windowsupdate.microsoft.com	Allow

Default rule:

Deny all outbound traffic.

Traffic Flow

1. VM initiates internet request.
2. Route table redirects traffic.
3. Azure Firewall receives traffic.
4. Firewall Policy evaluates FQDN.
5. Allowed traffic is forwarded.
6. Blocked traffic is denied and logged.

Expected Results

Allowed Access

- Azure portal services
- GitHub repositories
- Windows Updates

Blocked Access

- Social media websites
- Unknown domains
- Malware domains
- Unauthorized applications

Security Benefits

Centralized Policy Management

One firewall controls outbound access for multiple networks.

Reduced Attack Surface

Unauthorized outbound communication is blocked.

Improved Visibility

Administrators can analyze:

- Application access
- Threat indicators
- Policy violations

Consistent Enforcement

All virtual networks follow identical security rules.

Bottlenecks Students May Encounter

1. User-Defined Route Misconfiguration

Problem

Incorrect UDR configuration may:

- Block all internet access.
- Prevent Azure service connectivity.
- Cause application failures.

Symptoms

- VM cannot access allowed websites.
- DNS failures.
- Timeout errors.

Solution

Verify:

- Firewall private IP.
- Route table association.
- Next hop configuration.
- Effective routes.

2. Azure Firewall Cost

Problem

Azure Firewall has a significant hourly cost even with low traffic.

Impact

Long-running deployments may exceed educational budgets.

Cost Optimization

- Stop unused virtual machines.
- Delete firewall after testing.
- Use Azure student credits.
- Limit deployment duration.

Testing Scenarios

Test	Expected Result
Access github.com	Allowed
Access azure.microsoft.com	Allowed
Access windowsupdate.microsoft.com	Allowed
Access facebook.com	Blocked
Access random website	Blocked

Limitations

- Higher operational cost Centralized egress security
- FQDN-based filtering
- Simplified administration
- Comprehensive logging
- Scalable architecture
- Azure-native security integration
- Dependency on route configuration
- Firewall becomes a central point of failure if not designed with redundancy
- FQDN filtering primarily applies to HTTP/HTTPS traffic

Future Enhancements

- Azure Firewall Premium
- TLS inspection
- Threat intelligence filtering
- Web category filtering
- Multiple firewall zones
- Azure Firewall Manager
- SIEM integration with Microsoft Sentinel

Conclusion

The Azure Firewall Centralised Egress Filtering project demonstrates a practical enterprise cloud security architecture where all outbound traffic is routed through a centralized Azure Firewall. By enforcing an FQDN allowlist and blocking unauthorized destinations, the solution improves security, visibility, and policy consistency across multiple virtual networks. The project also highlights important operational challenges such as UDR misconfiguration and firewall cost, providing valuable experience in designing and managing secure Azure network environments.

References

1. Microsoft Azure Firewall Documentation
2. Azure Firewall Policy Documentation
3. Azure Virtual Network Documentation
4. Azure Route Table Documentation
5. Microsoft Learn – Secure Azure Networks